CO3-AT1

CSA0406 - OPERATING SYSTEMS

NAME:P.Jessica Hephzbah

REG NO:192511425

Q1) Base and Limit Register Address Translation

C-CODE & OUTPUT:

main.c

```c
#include <stdio.h>

int main()
{
    int base, limit, logical, physical;

    printf("Enter base register value: ");
    scanf("%d", &base);

    printf("Enter limit register value: ");
    scanf("%d", &limit);

    printf("Enter logical address: ");
    scanf("%d", &logical);

    if(logical < limit)
    {
        physical = base + logical;
        printf("Valid Address\n");
        printf("Physical Address = %d", physical);
    }
    else
    {
        printf("Invalid Address - Address Limit Exceeded");
    }

    return 0;
}
```

Output

```
Enter base register value: 1000
Enter limit register value: 500
Enter logical address: 200
Valid Address
Physical Address = 1200

=== Code Execution Successful ===
```

Q2) Paging Memory Management

C-CODE & OUTPUT:

main.c

```c
#include<stdio.h>

int main()
{
    int pageTable[10], pages, page, offset, size;

    printf("Enter number of pages: ");
    scanf("%d",&pages);

    printf("Enter page size: ");
    scanf("%d",&size);

    printf("Enter page table:\n");

    for(int i=0;i<pages;i++)
        scanf("%d",&pageTable[i]);

    printf("Enter page number and offset: ");
    scanf("%d%d",&page,&offset);

    if(page < pages)
        printf("Physical Address = %d",
        pageTable[page]*size+offset);
    else
        printf("Invalid Page");

    return 0;
}
```

Output

```
Enter number of pages: 4
Enter page size: 100
Enter page table:
5
2
8
6
Enter page number and offset: 1 20
Physical Address = 220

=== Code Execution Successful ===
```

Q3) Page Table Simulation

CODE & OUTPUT

main.c

```c
#include<stdio.h>

int main()
{
    int pageTable[5]={3,5,7,9,2};
    int size=100;
    int address,page,offset;

    printf("Enter logical address: ");
    scanf("%d",&address);

    page=address/size;
    offset=address%size;

    printf("Page Number = %d\n",page);
    printf("Frame Number = %d\n",pageTable[page]);

    printf("Physical Address = %d",
    pageTable[page]*size+offset);

    return 0;
}
```

Output

```
Enter logical address: 240
Page Number = 2
Frame Number = 7
Physical Address = 740

=== Code Execution Successful ===
```

Q4) TLB Simulation

CODE & OUTPUT

main.c

Share

Run

```c
#include<stdio.h>

int main()
{
    float hit,tlb,mem,emat;

    printf("Enter hit ratio: ");
    scanf("%f",&hit);

    printf("Enter TLB time and Memory time: ");
    scanf("%f%f",&tlb,&mem);

    emat = hit*(tlb+mem)+(1-hit)*(tlb+2*mem);

    printf("EMAT with TLB = %.2f",emat);

    printf("\nWithout TLB = %.2f",mem);

    return 0;
}
```

Output

```
Enter hit ratio: 0.8
Enter TLB time and Memory time: 10 100
EMAT with TLB = 130.00
Without TLB = 100.00

=== Code Execution Successful ===
```

Q5) Memory Protection using Base and Limit

CODE & OUTPUT:

main.c

```c
#include<stdio.h>

int main()
{
    int base,limit,address;

    printf("Enter base and limit: ");
    scanf("%d%d",&base,&limit);

    printf("Enter address: ");
    scanf("%d",&address);

    if(address>=base && address<=limit)
        printf("Access Allowed");
    else
        printf("Access Denied");

    return 0;
}
```

Output

```
Enter base and limit: 1000 1500
Enter address: 1499
Access Allowed

=== Code Execution Successful ===
```